BASICS OF GenAI

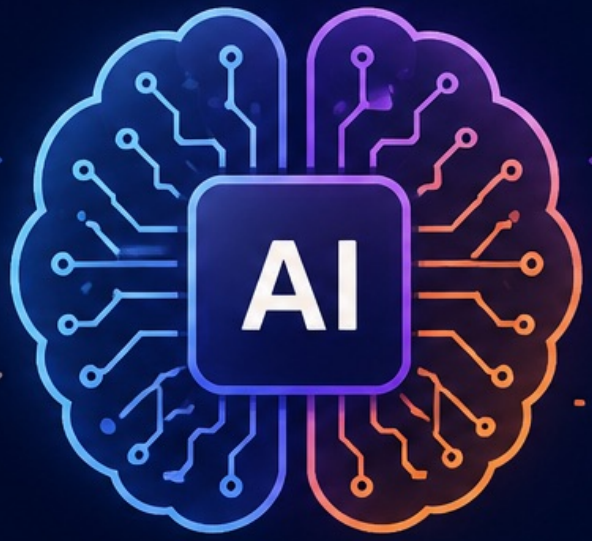
Understand. Create. Innovate.



TEXT GENERATION



IMAGE GENERATION



CODE GENERATION



IDEAS TO IMPACT





UNDERSTAND



CREATE



INNOVATE



EMPOWER

Lecture: Evaluation of LLMs

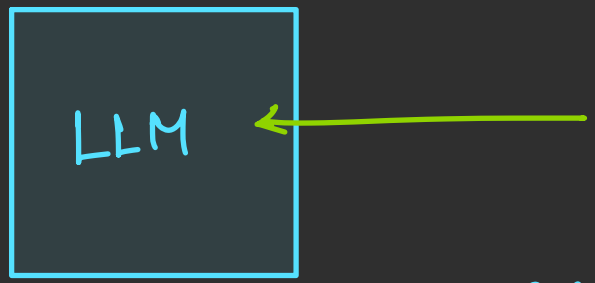
Agenda

1. OpenAI api basics
2. Understanding evaluations
3. Bertscore
4. LLM as a judge

#OpenAI API Basics

↳ LLM : GPT 5.4

~500 - 800 GB + Heavy RAM



servers owned by openAI

Evaluation

(LLM product)

→ AI product that uses an LLM

1) Customer support Chatbot
2) Code generation tool (Cursor/Co-pilot)
3) Wayfair (Sales Agent)

Types of LLM System

1) Single prompt task (Eg: Caption generator, Transcript generat)

Input → LLM → output

2) RAG System

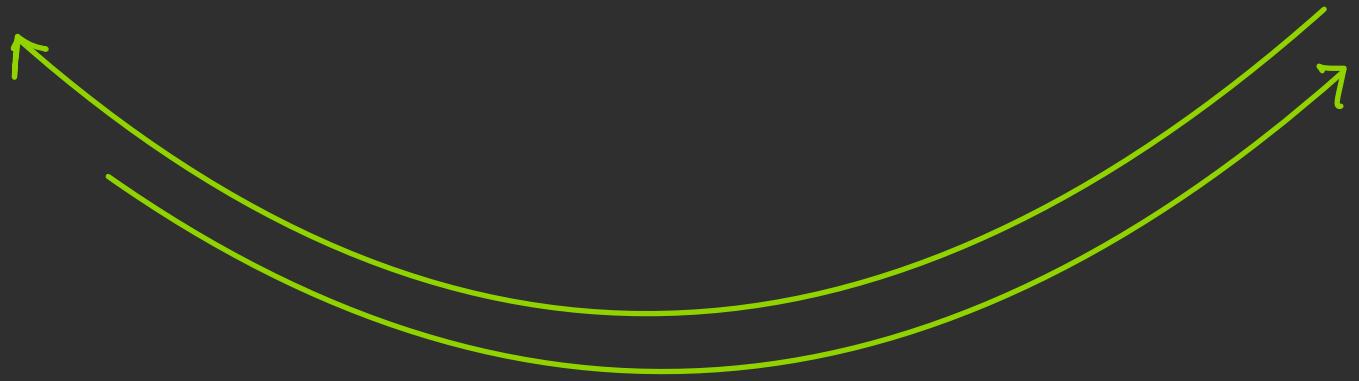
LLM with search/retrieval

KB

Chat bot with connected Information about my domain

3) Agentic System

↳ goal based + self analytic / correction + path finding



Notice

I am NOT available on 30th May as I have to travel, Hence the class on 30th May has been allocated to (Rohit Jindal) as a substitute.

[Note: Based on ops Team discussion]

Why Evaluation of LLMs are Hard?

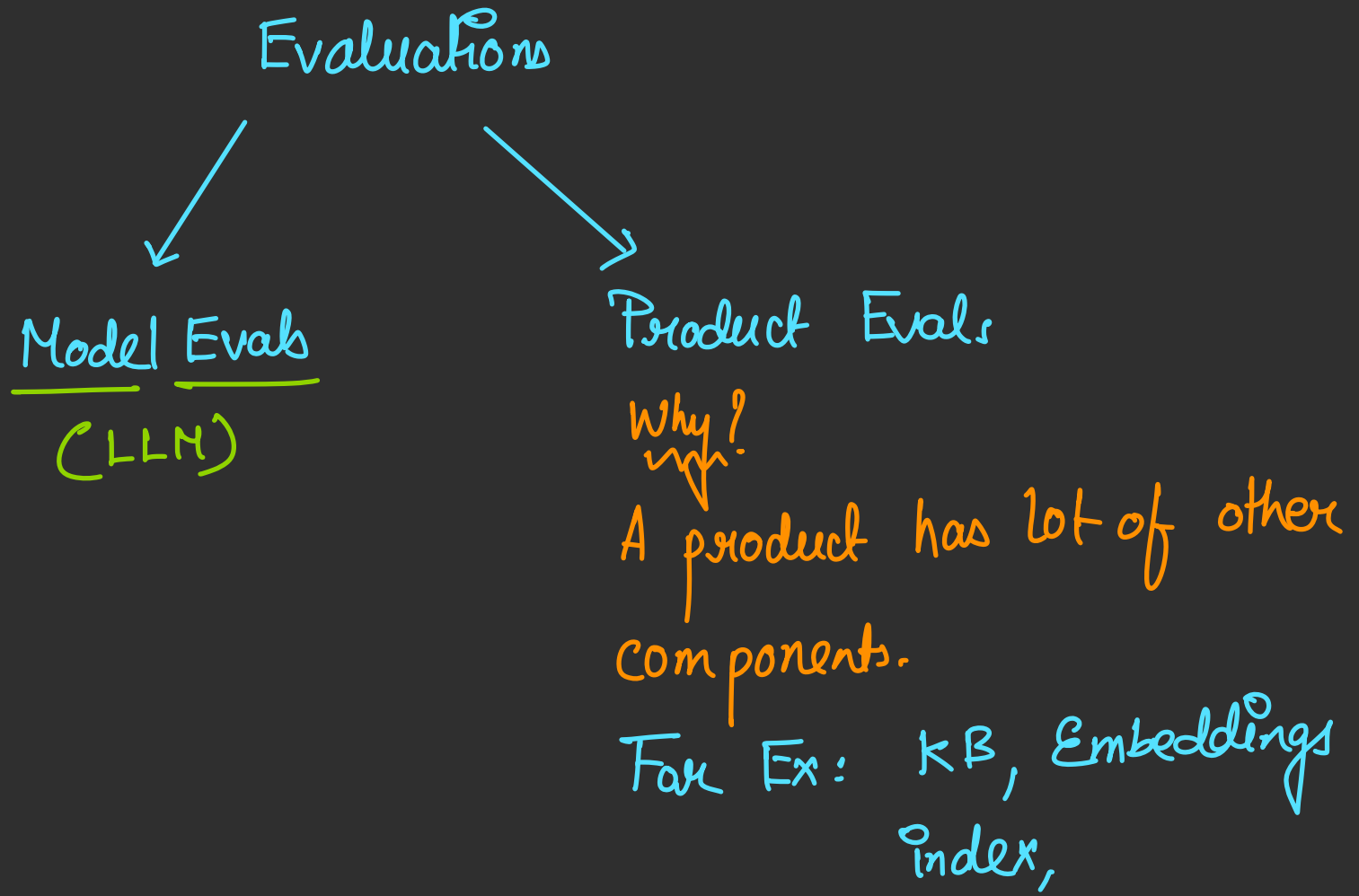
1) Non-deterministic Behaviour

same I/p → multiple output

2) No Single Correct Answer

↳ Multiple valid response is possible

3) Wide range of possible input


Evaluations
Model Evals
(LLM)
Product Evals
Why?
A product has lot of other components.
For Ex: KB, Embeddings index,

Bleu

↳ overlaps of n-grams

2-gram

Ref

He often gets school early

Candidate

He usually arrives early for class

overlap = 0

BERT Score

→ Capture Context using embedding

+

pairwise cosine similarity.

IDF

↳ Inverse Document Frequency

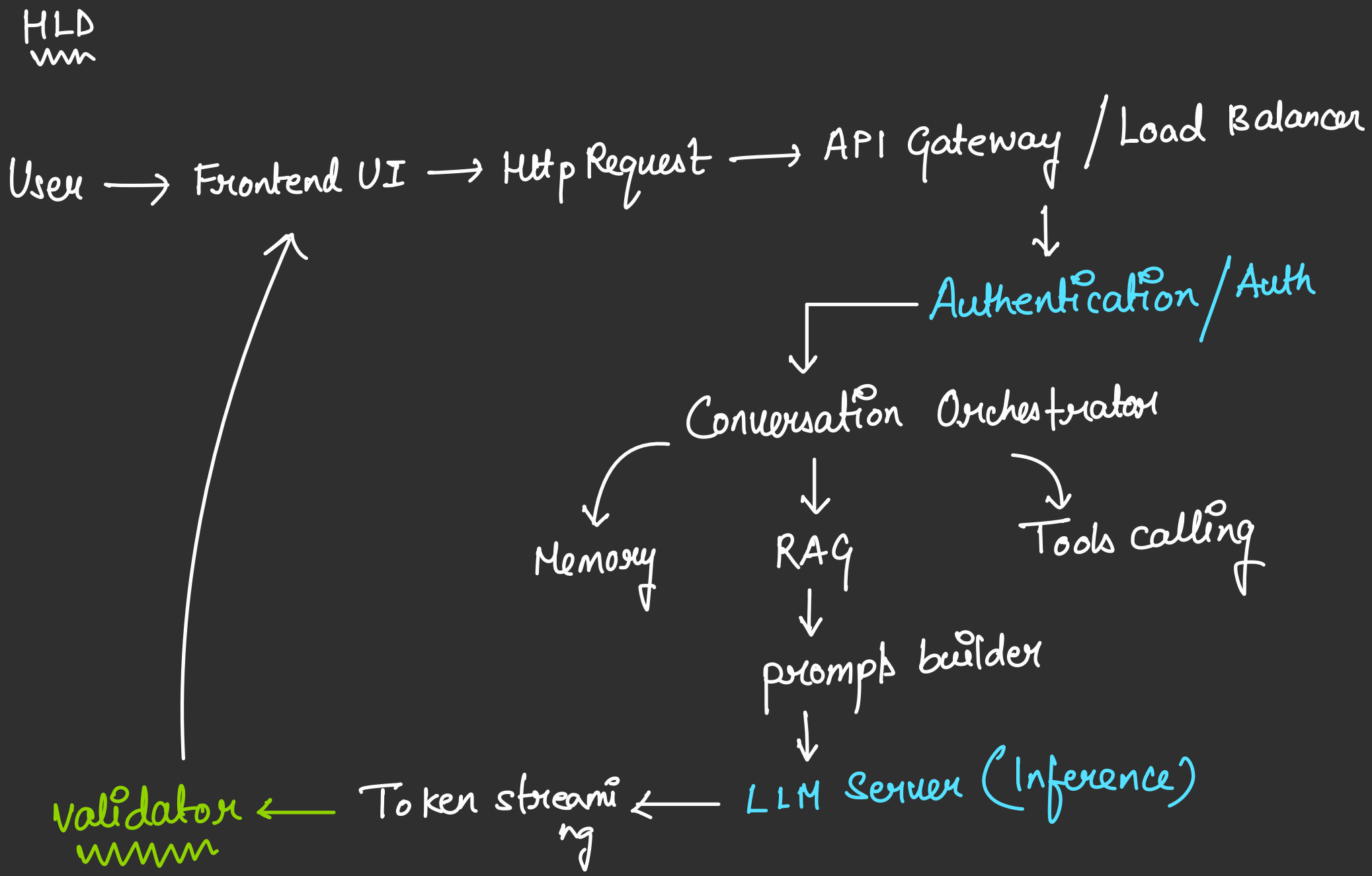
$$IDF(t) = \log \frac{N}{DF(t)}$$

↳ token

↳ Document frequency.

N → total Number of document in your corpus

DF(t) ⇒ No of document containing the token t


HLD
User
Frontend UI
Http Request
API Gateway / Load Balancer
Authentication / Auth
Conversation Orchestrator
Memory
RAG
Tools calling
prompts builder
LLM Server (Inference)
Token streaming
validator